

Nuclear Actin in Gene Regulation: From Transcriptional Mechanisms to Therapeutic Potential

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Abstract

This article synthesizes current knowledge on nuclear actin's multifaceted role in gene transcription, chromatin remodeling, and cellular reprogramming. Targeting researchers and drug development professionals, it explores foundational mechanisms of nuclear actin in RNA polymerase function, methodological advances for studying nuclear actin dynamics, troubleshooting for technical challenges, and validation through disease-relevant models. The review highlights nuclear actin as a central regulator of inducible transcription and discusses its emerging implications for understanding cancer biology and developing novel therapeutic strategies.

Nuclear Actin Fundamentals: Mechanisms of Transcriptional Control and Chromatin Remodeling

The paradigm of actin as solely a cytoskeletal protein has been fundamentally rewritten over the past six decades. Initially observed in the nucleus in the 1960s, nuclear actin was met with skepticism for nearly forty years until compelling molecular evidence in the early 2000s established its essential roles in transcription, chromatin remodeling, and gene regulation [1] [2]. This whitepaper traces the historical evolution of nuclear actin research, from its controversial beginnings to its current recognition as a master transcriptional regulator. We detail the key experimental breakthroughs that revealed nuclear actin's functions in all three RNA polymerase complexes, its critical partnership with actin-binding proteins and myosins, and its newly discovered roles in genome organization and cellular differentiation. Within the broader thesis of nuclear actin gene regulation research, this analysis provides a comprehensive technical resource

for scientists and drug development professionals seeking to understand and target actin-dependent nuclear processes.

The discovery of actin in the nucleus dates back more than sixty years, when initial observations detected its presence in nuclear compartments [1]. However, for the following forty years, the concept of functional nuclear actin received little research attention, largely due to the dominant paradigm of actin as a cytoplasmic structural protein and technical difficulties in discriminating nuclear actin from its highly abundant cytoplasmic counterpart [1] [2]. The turning point came around the year 2000, when a series of convincing experimental data emerged demonstrating that actin participates in essential nuclear processes [1]. This sparked a research renaissance that has established nuclear actin as a critical player in transcription, DNA repair, replication, chromatin remodeling, and nuclear organization [1].

The evolution of our understanding reflects a fundamental shift from viewing actin through a purely structural lens to recognizing it as a dynamic regulatory molecule that transduces signals from the cellular environment to the genome. This whitepaper examines key historical milestones in this field, providing technical details of foundational experiments and analyzing how each breakthrough expanded our understanding of nuclear actin's mechanistic roles in gene regulation.

Historical Timeline: Key Discoveries

Table 1: Historical Milestones in Nuclear Actin Research

Time Period	Key Discovery	Experimental Evidence	Significance
1960s-1970s	Initial observation of nuclear actin	Early biochemical and morphological studies [1] [2]	First suggestion of nuclear localization; largely ignored
1980s	Involvement in transcription	Antibody microinjection in salamander oocytes inhibited RNA synthesis [3] [4]	First functional evidence for nuclear role
1990s-2000s	Component of chromatin remodeling complexes	Identification in BAF complex binding to BRG1 ATPase subunit [3] [5]	Molecular mechanism for gene regulation
Early 2000s	Integral component of all RNA polymerases	Co-immunoprecipitation with RNA pol I, II, III; inhibition studies [3] [4]	Established fundamental role in basal transcription

Time Period	Key Discovery	Experimental Evidence	Significance
2000-2010	Role in transcriptional reprogramming	Nuclear transplant experiments in oocytes showing actin polymerization required for Oct4 reactivation [6]	Connected nuclear actin to cell fate determination
2010-2020	Mechanosensing and gene regulation	Actin-mediated regulation of MRTF-A/SRF pathway [7]	Linked cytoplasmic signaling to nuclear function
2015-Present	Genome organization and phase separation	Advanced imaging showing actin polymerization in Pol2 clusters [1] [5]	Revealed role in 3D genome architecture

Foundational Evidence: From Controversy to Acceptance

Early Resistance and Technical Challenges

The initial resistance to nuclear actin concepts stemmed from several factors. First, actin is highly abundant in the cytoplasm, making contamination a persistent concern in early biochemical preparations. Second, the conventional structural role of cytoplasmic F-actin created a conceptual barrier to imagining its nuclear functions. Third, visualization techniques were inadequate to definitively distinguish nuclear from cytoplasmic actin [2]. The field only gained momentum with the development of more sophisticated experimental tools, including improved fractionation methods, specific antibodies, and advanced imaging technologies [1].

Microinjection Experiments: The First Functional Evidence

The first compelling functional evidence emerged in the 1980s through microinjection experiments. When anti-actin antibodies were injected into the nuclei of salamander oocytes, researchers observed a dramatic contraction of the lateral loops of lamphrush chromosomes and complete inhibition of RNA synthesis [3] [4]. This seminal finding provided the first direct evidence that actin was not merely present in the nucleus but was functionally required for transcription. Similar experiments in HeLa cells further demonstrated that anti-actin antibodies inhibited pre-rRNA synthesis, establishing actin's role across cell types and RNA polymerases [3] [4].

Nuclear Actin in Transcription: Molecular Mechanisms

RNA Polymerase Complexes

A major breakthrough came in the early 2000s with the discovery that actin is a constitutive component of all three RNA polymerase complexes [3]. The evidence for this conclusion is summarized below:

Table 2: Evidence for Actin's Role in RNA Polymerase Complexes

RNA Polymerase	Key Evidence	Molecular Interactions	Functional Role
RNA Polymerase II	- β -actin in pre-initiation complexes [3]- Antibodies inhibit 15-nt transcript production [3]- Actin recruits Pol II to PICs [3]	Direct interaction with Pol II; TBP binding to TATA box [3]	PIC formation; initiation-elongation transition; recruitment
RNA Polymerase I	- Association with rDNA genes [3]- Anti-actin antibodies inhibit pre-rRNA synthesis [3]	Complex with nuclear myosin I (NMI); ATP-dependent stabilization [3]	Acts as molecular motor with myosin; facilitates elongation
RNA Polymerase III	- Direct interaction with RPC3, RPABC2, RPABC3 subunits [3]- Located at U6 gene promoter in vivo [3]	Photochemical cross-linking with DNA [3]	Essential for basal transcription; monomeric form required

Chromatin Remodeling Complexes

Concurrent with discoveries in transcription complexes, nuclear actin was identified as an integral component of major chromatin remodeling complexes. Nuclear β -actin was first identified in the BAF (BRG1/BRM-associated factor) chromatin remodeling complex, a SWI/SNF-like complex crucial for gene activation during T-lymphocyte activation [3] [5]. In this complex, actin binds directly to the BRG1 ATPase subunit and stimulates its ATPase activity, which is necessary for stable association of the complex with chromatin [3] [5].

Subsequent research revealed actin and actin-related proteins (ARPs) in multiple chromatin remodeling complexes including INO80, SWR1, NuA4, and TIP60 complexes [3] [2]. In the INO80 complex, actin is required for efficient DNA binding, ATPase activity, and nucleosome remodeling [3]. These complexes utilize actin and ARPs to regulate nucleosome positioning, histone modifications, and chromatin accessibility, thereby controlling gene expression patterns.

Figure 1: Nuclear Actin in Transcriptional Regulation. This diagram illustrates how extracellular signals influence cytoskeletal actin dynamics, which in turn regulates nuclear actin import and its

subsequent roles in chromatin remodeling, histone modification, and RNA polymerase function to control gene transcription.

Actin-Binding Proteins and Nuclear Function

The regulation of nuclear actin dynamics is controlled by numerous actin-binding proteins (ABPs) that control nucleation, bundling, filament capping, fragmentation, and monomer availability [3]. More than 30 actin-binding proteins have been discovered in nuclei, highlighting the sophisticated regulatory network controlling nuclear actin [2].

Key regulatory mechanisms include:

- **Nuclear Import/Export:** Nuclear actin levels are actively maintained by IMPORTIN9 (imports actin with cofilin) and EXPORTIN6 (exports actin as profilin-actin heterodimer) [2].
- **Polymerization Regulators:** Proteins like forming (mDia1/2), ARP2/3 complex, and N-WASp control nuclear actin polymerization in response to specific signals [1] [7].
- **Transcription Control:** STARS and ABLIM regulate actin dynamics and SRF-dependent muscle-specific gene expression [3].

The discovery and characterization of these regulatory mechanisms provided critical insights into how nuclear actin polymerization is spatially and temporally controlled to execute specific nuclear functions.

Experimental Approaches: Methodologies and Reagents

Key Experimental Protocols

Several experimental approaches have been pivotal in establishing nuclear actin functions:

Microinjection Assays: Early definitive experiments involved microinjection of anti-actin antibodies into nuclei of living cells (salamander oocytes, HeLa cells), followed by assessment of transcription inhibition through radiolabeled nucleotide incorporation or morphological changes in chromosome structure [3] [4].

Chromatin Immunoprecipitation (ChIP): Used to demonstrate actin recruitment to actively transcribed genes. Protocol involves: (1) formaldehyde cross-linking of proteins to DNA; (2) chromatin fragmentation by sonication; (3) immunoprecipitation with anti-actin antibodies; (4) reversal of cross-links and DNA purification; (5) PCR quantification of specific genomic regions [3].

Biochemical Fractionation and Complex Isolation: Nuclear extracts prepared through differential centrifugation and detergent treatments, followed by immunoprecipitation or affinity

purification of complexes. Critical for identifying actin in RNA polymerase and chromatin remodeling complexes [3] [5].

Live-Cell Imaging with Nuclear Actin Probes: Utilization of nuclear-targeted LifeAct or actin chromobody tags to visualize nuclear actin dynamics in response to stimuli like serum induction or mechanical stress [1] [7].

The Scientist's Toolkit: Essential Research Reagents

Table 3: Key Research Reagents for Nuclear Actin Studies

Reagent/Tool	Function/Application	Key Findings Enabled
Anti-actin antibodies	Microinjection to inhibit actin function	Established necessity for transcription [3]
CK666	ARP2/3 complex inhibitor; prevents actin branching	Revealed role in chromatin accessibility and stem cell differentiation [5]
Cytochalasin D	Inhibits cytoplasmic actin polymerization; increases nuclear actin	Demonstrated nuclear actin role in osteogenic differentiation [5]
Nuclear Actin Chromobody	Live-cell imaging of nuclear actin structures	Visualized nuclear actin polymerization dynamics [5]
LifeAct with NLS	Specifically labels filamentous actin in nucleus	Demonstrated serum-induced nuclear actin polymerization [7]
siRNA against β-actin	Specific knockdown of nuclear β-actin	Revealed roles in 3D genome organization [8]

Signaling Pathways: Mechanical and Biochemical Integration

Nuclear actin serves as a critical integrator of mechanical and biochemical signals, exemplified by its role in the MRTF-A/SRF pathway. This pathway directly links cytoskeletal dynamics to gene expression:

- **Cytoplasmic Regulation:** In resting cells, MRTF-A is sequestered in the cytoplasm through binding to G-actin via its RPEL domain, which sterically occludes its nuclear localization signal [7].

- **Signal Activation:** Stimuli that promote actin polymerization (e.g., serum stimulation, mechanical stress) reduce the G-actin pool, freeing MRTF-A for nuclear import [7].
- **Nuclear Function:** Once in the nucleus, MRTF-A partners with SRF to activate transcription of cytoskeletal and immediate early genes [7].
- **Nuclear Actin Feedback:** Nuclear actin polymerization further enhances MRTF-A/SRF activity, while monomeric nuclear actin promotes MRTF-A nuclear export and inhibits its activity [7].

Figure 2: MRTF-A/SRF Mechanotransduction Pathway. This diagram illustrates how extracellular signals such as serum stimulation or mechanical stress activate RhoA, leading to actin polymerization and subsequent nuclear import of MRTF-A, which partners with SRF to activate target gene transcription, with nuclear actin polymerization providing a positive feedback loop.

Contemporary Research and Therapeutic Implications

Emerging Frontiers

Recent research has expanded nuclear actin functions into new domains:

- **Genome Organization:** Nuclear actin regulates 3D genome architecture through compartment switching and promoter-enhancer interactions. β -actin depletion alters H3K27 acetylation and affects enhancer-dependent transcriptional regulation [8].
- **Phase Separation:** Actin polymerization drives cluster formation of RNA polymerase II and transcription factors, facilitating transcriptionally active biomolecular condensates [1].
- **DNA Repair and Replication:** Nuclear F-actin facilitates directed movement of double-strand breaks and stalled replication forks, with ARP2/3-mediated actin branching crucial for replication stress response [1].
- **Stem Cell Differentiation:** Nuclear actin polymerization states guide mesenchymal stem cell fate decisions, with different actin structures promoting osteogenic versus adipogenic lineages [5].

Therapeutic Opportunities

The growing understanding of nuclear actin functions opens novel therapeutic avenues:

- **Cancer Therapeutics:** Aberrant nuclear actin regulation observed in cancer suggests potential targeting opportunities, particularly in pathways controlling mechanosensitive gene expression [9].
- **Differentiation Therapies:** Manipulating nuclear actin states could direct stem cell differentiation for regenerative medicine applications [5].

- **Fibrosis Treatment:** Targeting MRTF-A/SRF pathway could ameliorate fibrosis driven by excessive matrix gene expression [7].

The historical journey of nuclear actin from cytoskeletal component to transcriptional regulator represents a fundamental evolution in our understanding of cellular regulation. What began as a controversial observation in the 1960s has matured into a sophisticated framework explaining how actin integrates mechanical and biochemical signals to control gene expression, genome organization, and cell fate. The experimental approaches refined over decades - from microinjection to modern live-cell imaging and genomic analyses - have collectively built a compelling case for nuclear actin as a master regulator of nuclear function.

As research continues to unravel the complexities of nuclear actin dynamics, particularly in the contexts of development, disease, and therapeutic intervention, this field promises to yield further insights into the fundamental mechanisms of gene regulation. For drug development professionals and researchers, nuclear actin pathways represent emerging opportunities for therapeutic intervention across a spectrum of diseases, from cancer to fibrosis to differentiation disorders.

Advanced Methodologies: Techniques for Visualizing and Manipulating Nuclear Actin in Live Cells

The actin cytoskeleton, long recognized for its structural and mechanical roles in the cytoplasm, plays an equally vital role within the nucleus, where it regulates fundamental processes including gene transcription, chromatin remodeling, and DNA repair [5]. Nuclear actin exists in a dynamic equilibrium between monomeric (G-actin) and polymeric (F-actin) states, with this balance serving as a key determinant of transcriptional outcomes and cell fate decisions [5]. The ability to visualize and quantify these dynamics in living cells has emerged as a cornerstone of modern cell biology, enabling researchers to decipher how nuclear actin structure governs gene expression programs in development, homeostasis, and disease.

The investigation of nuclear actin presents unique technical challenges. Unlike its cytoplasmic counterpart, nuclear F-actin rarely forms stable, phalloidin-positive structures under normal conditions, necessitating the development of specialized biosensors [5]. Furthermore, the dense nuclear environment demands probes with high specificity and sensitivity to avoid artifacts and mislocalization. This technical guide examines the current landscape of live-cell nuclear actin probes, with particular emphasis on the oil GV system and optimized actin biosensors, providing researchers with the methodological foundation needed to advance this rapidly evolving field.

Fluorescent tools for probing actin dynamics

Fundamental classes of fluorophores for live-cell imaging

The selection of appropriate fluorophores forms the basis of effective live-cell imaging strategies. Current technologies primarily utilize two platforms: **small-molecule dyes** and **fluorescent proteins (FPs)**, each offering distinct advantages and limitations for specific applications [10].

Small-molecule dyes such as silicon-rhodamine derivatives provide excellent brightness, photostability, and narrow emission bandwidths, making them ideal for prolonged imaging sessions and super-resolution techniques [10]. Recent innovations include near-infrared variants that minimize cellular autofluorescence and phototoxicity, along with "caged" compounds such as carbofluoresceins and carborhodamines that can be activated on demand [10]. However, these synthetic dyes often lack inherent molecular specificity and may require sophisticated conjugation strategies for targeted delivery.

Fluorescent proteins, being genetically encodable, circumvent the delivery challenges associated with synthetic dyes. Extensive protein engineering has yielded FPs with improved brightness, photostability, and maturation kinetics, including mNeonGreen, mRuby3, and FusionRed [10]. Notably, oxFPs (oxidized FPs) exhibit enhanced photostability through engineered disulfide bonds, while monomeric variants minimize the risk of perturbing native protein interactions [10]. For nuclear actin specifically, the Nuclear Actin Chromobody (TagGFP-labeled) has emerged as a valuable tool, consisting of a single-domain antibody fragment that recognizes endogenous actin with minimal perturbation [5].

Table 1: Comparison of Fluorophore Platforms for Nuclear Actin Imaging

Platform	Examples	Advantages	Disadvantages	Best Applications
Small-Molecule Dyes	Silicon-rhodamine, Carbofluoresceins	High brightness, Excellent photostability, Narrow bandwidth	Limited specificity, Cell permeability challenges	Super-resolution microscopy, Long-term tracking
Fluorescent Proteins	mNeonGreen, mRuby3, Nuclear Actin Chromobody	Genetic encoding, Molecular specificity, Minimal perturbation	Larger size, Slower maturation, Potential oligomerization	Endogenous tagging, Long-term expression studies
Nonnatural Amino Acids	Coumarin, Dansyl, Prodan derivatives	Small size, Minimal	Low incorporation efficiency, Context-	Specific domain labeling,

Platform	Examples	Advantages	Disadvantages	Best Applications
		perturbation, Site-specific incorporation	dependent variability	Conformational studies

Strategies for labeling molecules of interest in live cells

Multiple molecular strategies exist for targeting fluorophores to specific cellular compartments and proteins, each with varying implications for nuclear actin research.

Fluorescent protein fusions represent the most straightforward approach, with the protein of interest directly fused to an FP [10]. While convenient, this method can perturb the stability, localization, or function of the tagged protein, particularly for proteins that assemble into larger complexes [10]. Systematic studies indicate that approximately 80% of FP-tagged proteins localize correctly compared to endogenous proteins, though functional validation remains essential [10]. For actin specifically, FP fusions must be carefully validated as the bulky tag may alter polymerization kinetics or protein interactions.

Split-FP systems offer a promising alternative wherein a small beta strand of the FP is fused to the protein of interest, with the remainder expressed separately [10]. This approach minimizes structural perturbation and is particularly amenable to CRISPR/Cas-mediated tagging of endogenous loci [10]. Although not yet widely applied to nuclear actin, this strategy holds significant potential for monitoring endogenous actin dynamics with minimal perturbation.

Incorporation of fluorescent nonnatural amino acids enables direct integration of fluorophores into proteins via expanded genetic code systems [10]. This approach sidesteps the bulkiness of FP fusions, with studies demonstrating improved functional retention compared to FP tags in certain contexts [10]. However, incorporation efficiency remains variable and context-dependent, limiting widespread adoption.

The oil GV system for studying nuclear actin

System fundamentals and applications

The oil GV system represents an advanced experimental platform for investigating the molecular mechanisms of actin-dependent transcriptional regulation. This approach utilizes **cell-free extracts** combined with **actin polymerization modulators** to reconstitute nuclear processes in a controlled environment, enabling precise dissection of causality in actin-transcription coupling.

A key application of this system has been elucidating the role of actin polymerization in transcription factory formation. Research using this approach has demonstrated that serum stimulation induces the formation of enhanced transcription factories at serum response gene loci, characterized by increased density, activity, and duration [\[11\]](#). These enhanced factories require nuclear actin, which co-localizes with transcription sites and potentially functions as a dynamic scaffold through phase separation mechanisms [\[11\]](#).

Experimental protocols for the oil GV system

Protocol 1: Reconstitution of Actin-Dependent Transcription Factories

- **Materials Required:**

- Permeabilized cells or nuclear preparations
- Cytoplasmic extract (HeLa or other relevant cell lines)
- Energy regeneration system (ATP, creatine phosphate, creatine kinase)
- NTPs (ATP, GTP, CTP, UTP)
- Actin polymerization modulators (Jasplakinolide, Latrunculin B)
- Fluorescently labeled uridine analogs (EU-based Click chemistry)
- Nuclear Actin Chromobody or fluorescent phalloidin derivatives

- **Methodology:**

- Prepare permeabilized cells using digitonin treatment (40 µg/mL in cytoskeletal buffer for 5 minutes on ice)
- Wash to remove endogenous cytosolic components
- Incubate with cytoplasmic extract supplemented with energy mix (1mM ATP, 10mM creatine phosphate, 10 µg/mL creatine kinase) and 0.5mM of each NTP
- Add specific actin modulators: Jasplakinolide (100 nM) to stabilize F-actin or Latrunculin B (1 µM) to depolymerize actin filaments
- Incorporate EU (5-ethynyl uridine) at 1 mM for 30 minutes to label nascent RNA
- Fix with 4% PFA for 15 minutes and process for Click chemistry detection
- Image using super-resolution microscopy (STORM/PALM) to quantify transcription factory size and density

- **Key Readouts:**

- Transcription factory number per nucleus
- Factory size distribution and density
- Co-localization coefficient between actin filaments and transcription sites
- Nascent RNA production quantified by EU intensity

Protocol 2: Assessing Actin Phase Separation in Transcription Factories

- **Materials Required:**

- Recombinant actin ($\geq 90\%$ pure, lyophilized)
- RNA polymerase II (commercial source or purified)
- Transcription factors (specific to genes of interest)
- Crowding agent (PEG-8000 or Ficoll PM-400)
- Microfluidic chambers for imaging
- High-sensitivity EMCCD or sCMOS camera

- **Methodology:**

- Prepare reaction mixture containing 2 μM recombinant actin, 50 nM RNA polymerase II, and relevant transcription factors in transcription buffer
- Add crowding agent (10% PEG-8000) to mimic nuclear environment
- Load into microfluidic chambers to allow droplet formation
- Initiate polymerization with the addition of polymerization buffer (2 mM MgCl_2 , 100 mM KCl)
- Image droplet formation and dynamics every 30 seconds for 60 minutes
- Quantify fusion events and partition coefficients of transcriptional components

- **Key Readouts:**

- Droplet formation kinetics
- Partition coefficients of transcriptional components
- FRAP recovery half-time for assessing material properties
- Correlation between actin polymerization state and droplet properties

Optimized actin biosensors for live-cell imaging

Advanced biosensor designs for nuclear actin

Recent advances in biosensor engineering have yielded sophisticated tools specifically optimized for monitoring actin dynamics in the nuclear compartment. These include **FRET-based actin biosensors** that report on actin polymerization state through changes in energy transfer, **partitioning biosensors** that accumulate in specific actin structures, and **single-fluorophore biosensors** with environmental sensitivity.

The **Nuclear Actin Chromobody** represents one of the most widely utilized tools, consisting of a single-domain antibody fragment (nanobody) fused to a fluorescent protein [5]. This probe enables visualization of endogenous actin without overexpression, significantly minimizing perturbation of

native actin dynamics. When combined with CRISPR/Cas9-mediated tagging approaches, the chromobody strategy allows for monitoring actin dynamics at endogenous expression levels, providing a more physiologically relevant readout compared to overexpression systems.

Multiplexed biosensor approaches represent another frontier in nuclear actin research. Recent methodologies now enable concurrent tracking of numerous signaling activities through genetic barcoding systems spectrally separable from commonly used biosensors [12]. When combined with deep learning-based image analysis, this approach permits massively parallel observation of signaling networks, potentially revealing how nuclear actin dynamics coordinate with other biochemical activities in response to stimuli.

Quantitative comparison of nuclear actin biosensors

Table 2: Performance Characteristics of Nuclear Actin Biosensors

Biosensor Type	Molecular Design	Spatial Resolution	Temporal Resolution	Key Applications	Validation Requirements
Nuclear Actin Chromobody	GFP-tagged actin nanobody	Diffuse nuclear pattern, detects fibrils upon stimulation	Minutes to hours (limited by maturation)	Monitoring endogenous actin dynamics, Drug treatments	Compare with phalloidin in fixed cells, siRNA knockdown
FRET-Based Actin Biosensors	F-actin binding domain fused to FP pair	Subdiffraction limited through FRET index	Seconds to minutes	Monitoring rapid polymerization dynamics, Mechanical stimulation	Calibrate with actin drugs, Verify nuclear localization
Lifeact-Based Probes	17-aa peptide fused to FP	Limited by overexpression artifacts	Seconds to minutes	General actin dynamics (use with caution in nucleus)	Compare with chromobody, Potential actin bundling artifacts

Biosensor Type	Molecular Design	Spatial Resolution	Temporal Resolution	Key Applications	Validation Requirements
Actin Single FPs	Actin fused to monomeric FPs	Limited by overexpression	Minutes to hours	Long-term tracking, Expression studies	Verify functional competence, Localization compared to endogenous

Experimental protocols for biosensor validation and imaging

Protocol 3: Validating Nuclear Actin Biosensors with Pharmacological Modulators

- **Materials Required:**
 - Cells expressing nuclear actin biosensor
 - Serum-free culture medium
 - Actin drugs: Cytochalasin D (1-10 μ M), CK666 (50-100 μ M), Jasplakinolide (100 nM-1 μ M)
 - Live-cell imaging chamber with environmental control
 - Confocal or spinning disk microscope with 60-100x objective
- **Methodology:**
 - Culture cells expressing nuclear actin biosensor in glass-bottom dishes
 - Serum-starve for 4-12 hours to establish baseline
 - Pre-treat with actin drugs for 30-60 minutes before stimulation
 - Stimulate with serum (10-20%) or specific ligands (LPA, S1P)
 - Image every 30-60 seconds for 30-120 minutes
 - For FRET biosensors, acquire both donor and acceptor channels with appropriate controls
- **Key Readouts:**
 - Nuclear-to-cytoplasmic ratio of biosensor signal
 - FRET efficiency changes over time
 - Response amplitude and kinetics to stimuli
 - Biosensor localization relative to chromatin markers

Protocol 4: Multiparameter Imaging of Actin-Transcription Coupling

- **Materials Required:**

- Cells co-expressing nuclear actin biosensor and transcription reporter
 - EU or FUrD for nascent RNA labeling
 - Click chemistry reagents
 - High-content imaging system or confocal microscope
 - Image analysis software (ImageJ, CellProfiler, or commercial packages)
- **Methodology:**
 - Transfect or transduce cells with nuclear actin biosensor
 - Introduce transcription reporter (MS2 system, RNA stem loops)
 - Pulse-label with EU (1 mM) for 15-30 minutes
 - Live-cell imaging to capture actin dynamics and transcription site appearance
 - Fix and perform Click chemistry to detect nascent RNA
 - Correlate actin dynamics with transcription site formation
 - **Key Readouts:**
 - Temporal correlation between actin polymerization and transcription activation
 - Spatial association between actin structures and transcription factories
 - Statistical significance of actin-transcription coupling

The scientist's toolkit: Essential research reagents and materials

Table 3: Research Reagent Solutions for Nuclear Actin Studies

Reagent Category	Specific Examples	Function/Application	Key Considerations
Actin Polymerization Modulators	Cytochalasin D (1-10 μM), Jasplakinolide (100 nM-1 μM), CK666 (50-100 μM), Latrunculin A/B (1-5 μM)	Perturb actin dynamics to establish causality, CK666 inhibits Arp2/3 branching [5]	Cytochalasin D increases nuclear F-actin; CK666 decreases it [5]
Live-Cell Actin Probes	Nuclear Actin Chromobody (TagGFP), SiR-actin, Lifeact-GFP, F-tractin-TagRFP	Real-time visualization of actin dynamics in living cells	Chromobody recognizes endogenous actin; SiR-actin requires optimization of loading conditions

Reagent Category	Specific Examples	Function/Application	Key Considerations
Fixed-Cell Actin Stains	Fluorescent phalloidin conjugates (TRITC, Alexa Fluor variants)	High-resolution imaging of F-actin structures in fixed cells	Phalloidin preferentially labels F-actin; may not detect all nuclear actin forms [13] [5]
Nuclear Markers	Hoechst 33342, DAPI, SYTO dyes, Histone H2B-GFP	Demarcate nuclear boundaries and chromatin organization	Essential for defining nuclear compartment in imaging analysis
Transcriptional Reporters	EU/FUrd incorporation, MS2/MCP system, RNA Pol II mutants	Monitor transcriptional activity in parallel with actin dynamics	EU Click chemistry compatible with some fixation methods
Microscopy Systems	Spinning disk confocal, TIRF, Light-sheet, Super-resolution (PALM/STORM)	Image acquisition with appropriate speed and resolution	Spinning disk ideal for live-cell; super-resolution for structural details [14]

Nuclear actin in cellular signaling and mechanotransduction

The integration of nuclear actin dynamics with cellular signaling pathways represents a critical interface between extracellular cues and genomic responses. Multiple signaling cascades converge on nuclear actin, including those activated by **GPCR ligands** (lysophosphatidic acid, sphingosine-1-phosphate), **receptor tyrosine kinases** (VEGF, EGF, FGF receptors), and **integrin-mediated mechanosensing** [\[15\]](#). These pathways ultimately influence actin dynamics through Rho GTPase family members and their effectors, particularly the actin-related protein 2/3 (Arp2/3) complex and formin proteins [\[15\]](#).

The **MRTF-SRF signaling axis** serves as a paradigmatic example of nuclear actin-mediated transcription regulation. In this pathway, cytoplasmic G-actin sequesters MRTF (myocardin-related transcription factor), while polymerization liberates MRTF for nuclear translocation, where it activates SRF (serum response factor)-dependent transcription of genes encoding cytoskeletal components and regulators [\[15\]](#). This establishes a elegant feedback loop wherein actin dynamics directly control the expression of actin itself and its regulatory machinery.

Mechanical forces represent another crucial regulator of nuclear actin and transcription. External mechanical stimuli are transmitted to the nucleus through **LINC complexes** (Linker of Nucleoskeleton and Cytoskeleton), which physically connect cytoskeletal elements to the nuclear envelope [16] [17]. This force transmission can induce nuclear deformation and actin polymerization, subsequently altering chromatin organization and gene expression [16]. Recent research demonstrates that dynamic substrate topographies trigger actin- and vimentin-mediated nuclear mechanoprotection mechanisms, including histone modifications that minimize DNA damage [18].

The continuing development of live-cell nuclear actin probes, particularly the refinement of the oil GV system and optimized biosensors, is transforming our understanding of how nuclear architecture and actin dynamics coordinate gene regulatory programs. The emerging picture reveals nuclear actin as a central player in cellular mechanotransduction, epigenetic regulation, and transcriptional responses to diverse stimuli.

Future advancements in this field will likely focus on several key areas: (1) developing improved biosensors with reduced perturbation and enhanced signal-to-noise ratios; (2) creating multiplexed imaging platforms that simultaneously track actin dynamics alongside multiple signaling activities and transcriptional outputs; and (3) establishing standardized validation protocols to ensure physiological relevance of observations. As these tools become more sophisticated and accessible, they will undoubtedly uncover new dimensions of nuclear actin function in health and disease, potentially revealing novel therapeutic targets for conditions ranging from cancer to developmental disorders.

Technical Challenges and Optimization Strategies in Nuclear Actin Research

The isolation of high-quality nuclear fractions is a critical first step in the study of chromatin-bound complexes and nuclear processes, including the rapidly advancing field of nuclear actin research. Optimized nuclear extraction protocols enable researchers to investigate fundamental nuclear events such as gene transcription, DNA repair, replication, and chromatin remodeling—processes now known to involve actin and its associated proteins [1]. The integrity of the nuclear extract directly influences the reliability of downstream applications, from western blotting and chromatin immunoprecipitation to more specialized techniques quantifying protein occupancy on chromatin.

Recent advances have illuminated the complex roles of nuclear actin, which participates in essential nuclear processes through various forms and modifications. Actin is now recognized to function in transcription by all three RNA polymerases, influences replication fork dynamics and DNA repair mechanisms, and contributes to chromatin organization through its interaction with chromatin

remodeling complexes [\[1\]](#). These findings underscore the necessity of extraction protocols that preserve the native state, composition, and interactions of chromatin-bound complexes. This guide provides a comprehensive framework for optimizing pre-extraction methods to ensure the accurate analysis of such complexes within the context of a broader research program on nuclear actin and gene regulation.

Core Principles of Nuclear Extraction

Fundamental Objectives and Challenges

The primary goal of nuclear extraction is to isolate intact nuclei from cellular material, subsequently extracting nuclear proteins—including those tightly associated with chromatin—while maintaining their biological activity and interaction states. This requires a carefully balanced approach: the extraction must be strong enough to liberate nuclear components from their architectural constraints but gentle enough to prevent protein degradation or the disruption of native complexes.

A significant technical challenge lies in the compositional complexity of the nucleus. The nuclear compartment houses genomic DNA, various RNA species, and a diverse proteome, including structural proteins like lamins, regulatory factors, and enzymatic complexes. Research using cryo-electron tomography has revealed the intricate organization at the nuclear periphery, where chromatin interacts with the lamin meshwork in a highly structured manner [\[19\]](#). The mechanical integrity of this environment, maintained by A-type and B-type lamins, presents a substantial barrier to extraction that protocols must overcome [\[19\]](#). Furthermore, the extraction process must contend with nucleases and proteases released during cell disruption, which can rapidly degrade the very components of interest.

Critical Technical Considerations

Several technical factors must be optimized to ensure a successful nuclear preparation:

- **Cellular Disruption Efficiency:** The initial lysis step must completely disrupt the plasma membrane and cytoplasmic structures without compromising nuclear integrity. The choice of disruption method depends on the cell or tissue type. Cultured mammalian cells require relatively gentle lysis conditions, whereas plant cells or bacterial samples need more vigorous mechanical disruption [\[20\]](#).
- **Preservation of Nuclear Complexes:** The buffer composition is crucial for preserving protein-protein and protein-DNA interactions that exist in vivo. This includes maintaining appropriate ionic strength, pH, and the inclusion of protective agents like glycerol, as well as fresh protease and nuclease inhibitors to prevent sample degradation [\[21\]](#) [\[22\]](#).

- **Contaminant Removal:** A high-purity nuclear extract requires the effective separation of nuclear contents from cytoplasmic contaminants. This is typically achieved through differential centrifugation—low-speed spins to remove unlysed cells and debris, followed by higher-speed centrifugation to pellet nuclei [21] [22]. The use of detergent-based extraction buffers helps to solubilize nuclear membranes and release chromatin-bound complexes while keeping cytoplasmic contaminants in the supernatant.

Standard Nuclear Extraction Protocols

Comprehensive Protocol Workflow

The following diagram illustrates the generalized workflow for nuclear extraction, applicable to various cell types and tissue samples. The process involves a series of buffer-based steps for cellular lysis, nuclear purification, and final extraction of nuclear proteins.

Detailed Step-by-Step Methodology

The standard nuclear extraction protocol can be broken down into several critical phases, each requiring specific reagents and conditions:

- **Cell Harvesting and Washing:** Grow cells to 70-80% confluency (approximately $2-5 \times 10^6$ cells for a 100 mm plate). Remove growth medium and wash the cell monolayer twice with phosphate-buffered saline (PBS) to remove residual serum proteins. Scrape cells into PBS or detach using trypsin/EDTA, then collect them in a conical tube. Pellet cells by centrifugation at $1,000 \times g$ for 5 minutes at 4°C and carefully discard the supernatant [22].
- **Cellular Lysis and Cytoplasmic Extraction:** Resuspend the cell pellet in a freshly prepared, ice-cold hypotonic lysis buffer (e.g., $1\times$ NE1 buffer from EpiQuik kit or 10 mM HEPES-KOH pH 7.9, 1.5 mM $\text{Mg}(\text{OAc})_2$, 10 mM KOAc, 0.5 mM DTT, 0.2 mM PMSF). Use approximately 100 μL per 10^6 cells. Vigorously vortex the suspension and incubate on ice for 10 minutes to allow for swelling and lysis. Centrifuge the lysate at $12,000 \times g$ for 1 minute at 4°C . Following centrifugation, carefully remove and save the cytoplasmic supernatant if needed, or discard it, leaving the intact nuclear pellet [22].
- **Nuclear Extraction:** Add a high-salt extraction buffer (e.g., NE2 buffer or 20 mM HEPES-KOH pH 7.9, 1.4 M KOAc, 1.5 mM $\text{Mg}(\text{OAc})_2$, 0.2 mM EDTA, 25% glycerol, 0.5 mM DTT, 0.2 mM PMSF) to the nuclear pellet. The volume should be proportional to the pellet size (approximately 10 μL per 10^6 cells). The high salt concentration helps to dissociate proteins from chromatin and nuclear structures. Incubate the suspension on ice for 15 minutes, with intermittent vortexing every 3 minutes. For tougher nuclear preparations, brief sonication (3×10 seconds) can be applied to increase extraction efficiency [22].

- **Clarification and Storage:** Centrifuge the nuclear suspension at maximum speed ($\geq 14,000 \times g$) for 10 minutes at 4°C to pellet insoluble debris, including chromatin and nuclear structures. Carefully transfer the clear supernatant, which constitutes the nuclear extract, to a new pre-chilled tube. Quantify the protein concentration using a Bradford or BCA assay, aliquot the extract, and flash-freeze in liquid nitrogen before storing at -80°C. Avoid multiple freeze-thaw cycles to preserve protein integrity and activity [22].

Specialized Extraction Buffer Formulations

Table 1: Composition of Standard Nuclear Extraction Buffers

Buffer Component	Hypotonic Lysis Buffer	High-Salt Extraction Buffer	Function
Buffer Base	10 mM HEPES-KOH, pH 7.9 [21]	20 mM HEPES-KOH, pH 7.9 [21]	Maintains physiological pH
Salt	10-20 mM KOAc [21]	1.4 M KOAc [21]	Low ionic strength for cell swelling / High ionic strength for complex dissociation
Divalent Cations	1.5 mM Mg(OAc) ₂ [21]	1.5 mM Mg(OAc) ₂ [21]	Stabilizes nuclear structure and enzymes
Detergent	Optional (e.g., 0.1% Igepal CA-630) [23]	Not typically added	Aids in membrane disruption
Stabilizer	-	25% Glycerol [21]	Stabilizes protein structure
Reducing Agent	0.5 mM DTT (fresh) [21] [22]	0.5 mM DTT (fresh) [21] [22]	Prevents protein oxidation
Protease Inhibitors	0.2 mM PMSF and/or cocktail [21] [22]	0.2 mM PMSF and/or cocktail [21] [22]	Prevents proteolytic degradation

Advanced and Specialized Applications

Chromatin-Specific Protein Analysis

For studies focusing specifically on proteins tightly bound to chromatin, standard extraction protocols may be insufficient. Advanced techniques like the **Chromoflow protocol** combine biochemical fractionation with flow cytometry to enable quantitative analysis of chromatin-bound

proteins in single cells throughout the cell cycle [23]. This method is particularly valuable for investigating the dynamics of nuclear actin and its binding partners during processes like replication stress and DNA damage response.

The Chromoflow protocol involves a critical **detergent-based extraction step** prior to fixation. Cells are harvested and resuspended in a specialized extraction buffer containing 0.1% Igepal CA-630, 10 mM NaCl, 5 mM MgCl₂, 0.1 mM PMSF, and 10 mM phosphate buffer (pH 7.4). This buffer permeabilizes the plasma membrane and solubilizes cytoplasmic components while leaving the nucleus and associated chromatin structures intact. The optimal detergent concentration is cell-type-dependent and must be determined empirically [23]. Following extraction, cells are fixed with formaldehyde, which cross-links and preserves chromatin-bound complexes. The fixed, permeabilized cells can then be immunostained for target proteins (e.g., actin, cohesin subunits, or DNA repair factors) and analyzed by flow cytometry. This approach allows for high-throughput, quantitative assessment of protein occupancy on chromatin and its correlation with cell cycle stage.

Preparation of Splicing-Competent Nuclear Extracts

Functional assays, such as in vitro splicing, require extracts that preserve not only the composition but also the enzymatic activity of macromolecular complexes. The preparation of splicing-competent nuclear extracts involves a more extensive subcellular fractionation process. The Dignam protocol and its variants begin with swollen cells that are gently dounced in a hypotonic buffer to release nuclei [21]. Nuclei are then subjected to a salting-out extraction using buffers containing 0.42 M KCl or higher to solubilize spliceosomal components and other nuclear machinery. The extract is dialyzed against a physiological buffer to restore salt concentrations compatible with enzymatic activity. The resulting nuclear extract supports complex biochemical reactions, including pre-mRNA splicing, making it an invaluable tool for dissecting the mechanistic roles of nuclear actin in transcription and RNA processing [21] [1].

Nucleic Acid-Free Nuclear Extracts

In applications where DNA or RNA contamination interferes with downstream analysis (e.g., DNA-protein binding assays like EMSA, or certain enzymatic assays), nucleic acid-free nuclear extracts are essential. Kits such as the EpiQuik Nuclear Extraction Kit II (Nucleic Acid-Free) incorporate specific enzymatic or precipitation-based clean-up steps to remove nucleic acids after the initial nuclear extraction [22]. This ensures that observed effects or interactions are attributable to proteins and not confounded by the presence of genomic DNA or RNA.

Research Reagent Solutions for Nuclear Studies

Table 2: Essential Reagents for Nuclear Extraction and Chromatin Analysis

Reagent / Kit	Primary Function	Key Features / Applications
EpiQuik Nuclear Extraction Kit [22]	Standard nuclear protein isolation	Complete 60-minute protocol; suitable for western blot, enzyme assays
EpiQuik Nuclear Extraction Kit II (Nucleic Acid-Free) [22]	Isolation of nuclear proteins free of nucleic acids	Removes contaminating DNA/RNA; ideal for protein-DNA binding studies
Igepal CA-630 [23]	Non-ionic detergent for membrane permeabilization	Critical for Chromoflow protocol; concentration must be cell-type optimized
Dithiothreitol (DTT) [21] [22]	Reducing agent	Prevents oxidation of protein thiol groups; must be prepared fresh
Protease Inhibitor Cocktail (PIC) [21] [22]	Inhibition of serine, cysteine, and metalloproteases	Prevents protein degradation during extraction; added to buffers before use
RNase A [20]	Ribonuclease A	Digests contaminating RNA in DNA-focused extractions or for nucleic acid-free extracts
Phenylmethanesulfonyl fluoride (PMSF) [21]	Serine protease inhibitor	Rapidly inactivates serine proteases; unstable in aqueous solution

Analytical and Functional Validation Methods

Quality Assessment of Nuclear Extracts

Validating the quality and purity of nuclear extracts is crucial for interpreting experimental results. Several methods can be employed:

- **Purity Analysis:** Western blotting is the most common technique for assessing extract purity. Blots should be probed for definitive nuclear markers (e.g., lamin A/C, histone H3, or transcription factors like RNA Polymerase II) and the absence of cytoplasmic contaminants

(e.g., GAPDH, tubulin, or cytochrome c). A high-quality extract will show strong signals for nuclear markers with minimal to no detection of cytoplasmic proteins [\[19\]](#) [\[22\]](#).

- **Functional Assays:** The ultimate validation of a nuclear extract is its performance in downstream applications. For splicing-competent extracts, this involves demonstrating accurate and efficient in vitro splicing of a radiolabeled pre-mRNA substrate, followed by visualization of the spliced products via denaturing gel electrophoresis and autoradiography [\[21\]](#). For extracts intended for chromatin studies, functional competence can be assessed by their ability to support chromatin remodeling in reconstituted systems or to exhibit expected protein-DNA interactions in EMSA experiments.

Integration with Advanced Genomic and Structural Techniques

Nuclear extracts serve as the starting material for a wide range of sophisticated analyses that probe chromatin architecture and function:

- **Chromatin Conformation Analysis:** Techniques such as Chromatin Interaction Analysis by Paired-End Tag Sequencing (ChIA-PET) and high-throughput chromosome conformation capture (Hi-C) rely on cross-linked chromatin, often prepared from intact nuclei, to map the three-dimensional organization of the genome. These methods have revealed how chromatin loops bring distal regulatory elements, such as enhancers, into proximity with target gene promoters, thereby influencing transcriptional regulation [\[24\]](#).
- **Structural Studies:** Cryo-electron tomography (cryo-ET) of cryo-focused ion beam (cryo-FIB) milled cells provides nanometer-scale resolution of native nuclear structures. This technology has been instrumental in visualizing the direct interactions between nucleosomes and the nuclear lamina meshwork, revealing how different lamin isoforms (A/C vs. B-type) distinctly influence chromatin organization and density at the nuclear periphery [\[19\]](#). Such structural insights are fundamental to understanding the mechanical and regulatory roles of the nucleus.

Troubleshooting and Optimization Strategies

Even with standardized protocols, researchers may encounter challenges during nuclear extraction. The following table addresses common issues and provides evidence-based solutions.

Table 3: Troubleshooting Common Nuclear Extraction Problems

Problem	Potential Cause	Optimization Strategy
Low Yield	Inefficient cell lysis or nuclear extraction	Optimize detergent concentration and type [23] ; Increase incubation time with

Problem	Potential Cause	Optimization Strategy
		extraction buffer; Incorporate brief sonication pulses [22]
Cytoplasmic Contamination	Incomplete lysis or overly harsh lysis damaging nuclei	Titrate detergent in lysis buffer (0.1%-0.5%) [23]; Verify lysis efficiency microscopically; Avoid vortexing during lysis step
Proteolytic Degradation	Inadequate inhibition of proteases	Ensure protease inhibitors are fresh and added to all buffers [21] [22]; Perform all steps at 4°C; Work rapidly
Poor Downstream Performance	Compromised protein activity or incorrect buffer composition	Use fresh DTT [21]; Avoid excessive salt concentrations unless required; Dialyze extract into appropriate buffer for functional assay [21]
Inconsistent Results Across Cell Lines	Cell-type specific differences in nuclear envelope stability	Empirically determine the ideal detergent concentration for each new cell type [23]; Adjust buffer salt composition based on cell origin

The field of nuclear analysis continues to advance with new methodologies for single-cell chromatin analysis [25] and sophisticated structural techniques [19] providing unprecedented insights. Optimized nuclear extraction remains the foundational step that enables these powerful applications, driving discovery in nuclear actin biology and gene regulatory mechanisms.

Short-term versus Long-term Depletion Approaches to Separate Interphase from Mitotic Functions

Many essential proteins perform distinct functions during **interphase** and **mitosis**. During interphase, which can last up to 20 hours in cycling cells, proteins often regulate fundamental nuclear processes such as gene transcription, chromatin organization, and DNA repair [26]. Conversely, during the relatively brief mitotic phase (approximately 40-60 minutes), the same proteins can be repurposed for functions such as spindle assembly, chromosome segregation, and cytokinetic regulation [26]. This functional duality presents a significant methodological challenge: how to experimentally dissect a protein's mitotic role from its interphase function. Traditional long-term depletion or knockout strategies inevitably affect both phases, confounding phenotypic interpretation. This technical guide, framed within the context of **nuclear actin gene regulation transcription research**, outlines strategic depletion approaches to resolve these temporally

distinct functions, providing methodologies critical for researchers and drug development professionals targeting specific cell cycle phases.

Conceptual Framework: Principles of Temporal-Specific Depletion

The core principle underlying temporal-specific depletion is the strategic manipulation of the protein loss-of-function timeline relative to cell cycle progression. The choice between short-term and long-term approaches depends on the protein's stability, turnover kinetics, and the specific biological question.

- **Long-Term Depletion:** This approach involves depleting a target protein for a duration exceeding one full cell cycle (typically >24 hours). It ensures near-complete protein loss in all cells but simultaneously perturbs both interphase and mitotic functions. Phenotypes observed are therefore composite, encompassing **direct mitotic defects** (e.g., failed contractile ring assembly) and **indirect consequences** of disrupted interphase processes (e.g., altered transcription of mitotic regulators) [26].
- **Short-Term/Acute Depletion:** This strategy aims to deplete a protein immediately before or during the phase of interest. By limiting the depletion window to less than one cell cycle, it is possible to study mitotic functions before significant interphase defects manifest, or vice versa. This is particularly powerful for studying proteins like Anillin (ANLN), where short-term depletion allows assessment of its nuclear role in transcription before the onset of mitotic failure and multinucleation [26].

Table 1: Comparison of Depletion Strategy Outcomes for the Oncoprotein ANLN

Depletion Strategy	Duration	Key Phenotypes Observed	Functional Phase Assessed
Long-Term Depletion [26]	>24 hours (e.g., 48-72 hrs)	Multinucleation, mitotic failure, contractile ring defects, and altered gene expression.	Composite phenotype from disrupted interphase and mitotic functions.
Short-Term Depletion [26]	<16 hours	Specific loss of RNA Polymerase II clustering at super-enhancers; altered chromatin binding; changes in transcription of immediate target genes.	Isolated interphase function in transcriptional regulation, prior to mitotic failure.

Technical Approaches for Acute Depletion

Several advanced molecular techniques enable the acute and rapid protein depletion required for temporal separation of function.

Auxin-Inducible Degron (AID) System

The AID system facilitates rapid, conditional protein degradation. Cells are engineered to express the target protein fused to an AID tag and the TIR1 E3 ubiquitin ligase. Adding auxin induces **ubiquitin-mediated degradation** of the tagged protein within minutes.

- **Protocol for Mitotic Exit Studies** [27]:
 - **Cell Line:** Use murine erythroblast (G1E-ER4) or human DLD-1 cells with endogenous AID-tagging of the target protein (e.g., CTCF, TRIP13) and stable TIR1 expression.
 - **Synchronization:** Arrest cells in prometaphase using a 16-hour treatment with 100 ng/mL nocodazole.
 - **Acute Depletion:** Release cells from the nocodazole arrest and immediately add 500 μ M indole-3-acetic acid (IAA, auxin) to the medium.
 - **Analysis:** Harvest cells at specific time points during mitotic exit (ana/telophase, early G1) for Hi-C, RNA-seq, or immunofluorescence. Degradation efficiency should be confirmed by western blotting [28] [27].
- **Key Findings Using This Approach:**
 - **CTCF Depletion:** Acute degradation during mitotic exit revealed that CTCF is required for the re-establishment of specific chromatin architectural loops and boundaries in nascent G1 nuclei, but not for global compartmentalization [27].
 - **TRIP13 Depletion:** Rapid degradation of TRIP13 in interphase caused a failure in mitotic checkpoint activation in the subsequent mitosis. In contrast, depletion after mitotic entry resulted in a prolonged checkpoint arrest, revealing its dual role in checkpoint activation and silencing [28].

Short-Term RNAi and Small-Molecule Inhibition

For proteins that are not amenable to genetic tagging, **transient siRNA transfection** with a carefully optimized short timeframe can achieve partial yet specific depletion.

- **Protocol for Separating Nuclear ANLN Function** [26]:
 - **Transfection:** Transfer HeLa or KYSE150 cells with ANLN-specific siRNA.
 - **Short-Term Window:** Analyze phenotypes 16-20 hours post-transfection. This window is short enough to prevent the multinucleation seen after 48-hour depletion.
 - **Phenotypic Analysis:** Use super-resolution microscopy (e.g., Structured Illumination Microscopy, SIM) to quantify RNA Polymerase II-pS5 clusters. Short-term depletion

specifically reduces large, transcriptionally active Pol II type III clusters without affecting global Pol II protein levels [26].

The Scientist's Toolkit: Essential Reagents for Depletion Studies

Table 2: Key Research Reagent Solutions for Temporal Depletion Studies

Reagent / Tool	Function in Experiment	Example Application
Auxin-Inducible Degron (AID) [28] [27]	Enables rapid, conditional protein degradation (t1/2 ~7 min).	Acute depletion of CTCF or TRIP13 during specific cell cycle stages [28] [27].
siRNA / shRNA [26]	Mediates transcript knockdown; timing controlled by transfection duration.	Short-term (16-20h) vs. long-term (48h) depletion of ANLN [26].
Nocodazole [29] [27]	Microtubule polymerization inhibitor; synchronizes cells in prometaphase.	Cell cycle synchronization for studies on mitotic exit and post-mitotic chromatin re-formation [29] [27].
2-Deoxyglucose (2-DG) & Sodium Azide (NaN3) [29] [30]	Metabolic inhibitors that deplete cellular ATP pools.	Inducing mitotic slippage during nocodazole arrest to study checkpoint adaptation [29] [30].
1,6-Hexanediol [26]	Alcoholic reagent that disrupts weak hydrophobic interactions.	Testing liquid-liquid phase separation (LLPS) properties of nuclear condensates like Pol II clusters [26].
Super-Resolution Microscopy (SIM) [26]	High-resolution imaging (~60 nm) to visualize subnuclear structures.	Quantifying changes in Pol II cluster size and number after ANLN depletion [26].
CUT&Tag / Hi-C [26] [27]	Maps genome-wide protein-DNA interactions and 3D chromatin architecture.	Assessing changes in Pol II chromatin binding and chromatin looping after acute CTCF loss [26] [27].

Experimental Workflow: From Depletion to Analysis

A robust experimental design for separating interphase and mitotic functions integrates depletion strategies with precise cell cycle monitoring and phase-specific readouts.

Data Interpretation and Mitigation of Confounding Factors

Accurately interpreting data from depletion studies requires careful consideration of several potential confounders.

- **Phenotypic Lag vs. Direct Effect:** A phenotype observed after short-term depletion is more likely to be a direct consequence of the protein's loss in that phase. For example, the specific reduction of Pol II clusters after short-term ANLN knockdown is a direct interphase effect [26]. In contrast, multinucleation after long-term depletion is an indirect consequence of failed cytokinesis in a previous mitosis.
- **Checkpoint Adaptation and Mitotic Slippage:** Under prolonged mitotic arrest (e.g., with nocodazole), cells can undergo **mitotic slippage**, exiting mitosis without dividing and becoming tetraploid G1 cells [29] [30]. This process can be influenced by experimental conditions; for instance, ATP depletion during nocodazole arrest promotes slippage via an APC/CCdh1-dependent pathway, rather than the canonical APC/CCdc20 pathway [29] [30]. This alternative exit mechanism must be considered when interpreting phenotypes from long-term mitotic arrests.
- **Architectural Memory from Mitosis:** The orderly progression through mitosis is critical for establishing interphase genome architecture. Depleting mitotic regulators (e.g., condensins, kinetochore proteins) can lead to aberrant interphase nuclear organization, such as altered centromere positioning, in the subsequent cell cycle [31]. This demonstrates that some "interphase" phenotypes are actually indirect results of **disordered mitosis**.

The strategic application of short-term versus long-term depletion approaches is paramount for deconvoluting the complex, phase-specific functionalities of cellular proteins. Techniques like the AID system and optimized short-term RNAi, combined with precise cell cycle synchronization and phase-appropriate analytical readouts, provide a powerful framework for this purpose. Within the field of nuclear actin research, these methods are indispensable for distinguishing actin's direct roles in interphase transcription—through mechanisms like Pol II clustering and chromatin remodeling [1]—from its structural functions in mitosis. For drug development, this temporal resolution helps in understanding on-target effects and identifying potential cycle-specific therapeutic windows, ultimately leading to more precise and effective interventions.

Validation in Disease Models and Comparative Analysis Across Biological Contexts

The nucleus, once considered a passive repository for genetic material, is now understood as a dynamic organelle whose structural and functional integrity is critical for proper gene regulation. Central to this understanding is the role of nuclear actin, which has emerged as a master regulator of essential nuclear processes including transcription, DNA repair, chromatin remodeling, and three-dimensional genome organization. This technical review examines how dysregulation of nuclear actin-mediated mechanisms contributes to disease pathogenesis, with particular emphasis on cancer progression and developmental disorders. We synthesize recent advances demonstrating that perturbations in nuclear actin dynamics disrupt transcriptional programs, DNA damage response pathways, and higher-order chromatin architecture, ultimately driving malignant transformation and developmental abnormalities. The therapeutic implications of targeting nuclear actin-dependent regulatory networks are also discussed, providing a framework for novel intervention strategies in human disease.

Actin is one of the most abundant and evolutionarily conserved proteins in eukaryotes, with roles extending far beyond its well-established cytoplasmic functions in cell shape, movement, and division. Over the past two decades, compelling evidence has established that actin is present in the nucleus and participates in essential nuclear processes [1]. Today, we understand that nuclear actin is involved in transcription, replication, DNA repair, chromatin remodeling, and contributes to the determination of nuclear shape and size [1] [32].

The conformational plasticity of actin and its ability to undergo regulated changes in polymerization states underlie its functional versatility in the nucleus. Both monomeric (G-actin) and polymeric (F-actin) forms exist within the nucleus, with transitions between these states facilitating distinct nuclear functions [32]. These dynamics are regulated by actin-binding proteins (ABPs), many of which shuttle between cytoplasmic and nuclear compartments in response to specific stimuli. Nuclear actin is subjected to numerous post-translational modifications—including acetylation, oxidation, phosphorylation, and SUMOylation—that influence its interactions with other proteins and its functional properties [1].

This review explores how disruptions in nuclear actin-mediated processes contribute to disease pathogenesis, with particular emphasis on cancer and developmental disorders. We examine the molecular mechanisms linking nuclear actin dysregulation to pathological gene expression, provide detailed experimental protocols for investigating these connections, and discuss emerging therapeutic opportunities targeting nuclear actin networks.

Nuclear Actin in Transcriptional Regulation and Disease

Fundamental Mechanisms of Actin-Dependent Transcription

Nuclear actin plays multifaceted roles in transcription by all three RNA polymerases. Early studies demonstrated that actin co-purifies with RNA polymerase II (Pol II) and is required for transcription initiation [32]. Subsequent work has revealed that actin participates in various phases of the transcription cycle, including pre-initiation complex assembly, transcription elongation, and co-transcriptional RNA processing [1] [32].

The polymerization state of actin determines its specific transcriptional functions. Monomeric actin regulates transcription factors such as the serum response factor (SRF) coactivator MAL/MRTF-A, while polymeric actin facilitates the formation of transcriptionally active biomolecular condensates [1] [33]. Recent research has identified stimulus-induced nuclear actin polymerization as a critical mechanism for enhancing RNA polymerase II clustering in a phase-separated state to promote transcription [1] [33]. For example, in response to serum stimulation, N-WASp/ARP2/3-mediated nuclear actin polymerization drives Pol II cluster formation [1]. Similarly, testosterone stimulation induces DAAM2 formin-dependent nuclear F-actin assembly that facilitates androgen receptor droplet formation and target gene expression [1] [33].

Myosin motor proteins complement actin's transcriptional functions. Nuclear myosin I (NMI) and myosin VI have been implicated in transcription, with NMI forming complexes with p53 to activate p21 expression upon DNA damage, and myosin VI serving as a molecular anchor that holds Pol II in high-density clusters [1].

Transcriptional Dysregulation in Disease

Dysregulation of actin-dependent transcriptional mechanisms contributes significantly to disease pathogenesis, particularly in cancer. In prostate cancer, nuclear F-actin drives testosterone-stimulated gene expression by promoting the formation of transcriptionally active condensates containing androgen receptors, DAAM2 formin, and active Pol II [33]. This mechanism enhances the expression of prostate-specific antigen and other oncogenic drivers.

The long non-coding RNA Meg3 provides another compelling example connecting nuclear actin dysfunction to disease. In β -actin depleted cells, Meg3 becomes enriched and binds to H3K27 acetylation marks within gene regulatory regions, disrupting promoter-enhancer interactions and repressing genes involved in chondroitin, heparan, dermatan sulfate, and phospholipase biosynthetic pathways [8]. These findings suggest that nuclear actin depletion—a common feature in some cancers—can trigger extensive genetic reprogramming through lncRNA-mediated mechanisms.

Table 1: Nuclear Actin-Related Transcriptional Mechanisms in Disease

Disease Context	Nuclear Actin Mechanism	Transcriptional Outcome	Reference
Prostate Cancer	DAAM2 formin-mediated nuclear F-actin assembly	Enhanced androgen receptor signaling and target gene expression	[1] [33]
Multiple Cancers	β -actin depletion-induced Meg3 lncRNA upregulation	Disrupted promoter-enhancer interactions and metabolic pathway repression	[8]
DNA Damage-Associated Diseases	NMI-p53 complex formation at damage response genes	Altered expression of cell cycle regulators like p21	[1]

Nuclear Actin in Chromatin Organization and Genome Stability

Actin-Dependent Chromatin Remodeling and 3D Genome Organization

Nuclear actin is an integral component of several chromatin remodeling complexes, including SWI/SNF/BAF, INO80, and NuA4 [\[1\]](#) [\[8\]](#). These complexes utilize the mechanical properties of actin to reposition nucleosomes and modulate DNA-histone interactions, thereby regulating chromatin accessibility. The role of actin in chromatin remodeling complexes extends beyond structural support; specific post-translational modifications on actin, such as arginine monomethylation at R256 in yeast actin, influence its function within these complexes [\[1\]](#).

Beyond nucleosome-level changes, nuclear actin contributes to higher-order chromatin organization. Depletion of β -actin leads to significant compartment switching, where chromatin regions shift between active (A) and inactive (B) compartments [\[8\]](#). These alterations in 3D genome architecture impact chromatin remodeling activities and nuclear compartmentalization. In β -actin knockout mouse embryonic fibroblasts (MEFs), the Polycomb Repressive Complex 2 (PRC2) replaces SWI/SNF, leading to increased methylation, chromatin compaction, and disrupted enhancer-dependent transcriptional regulation [\[8\]](#).

DNA Repair and Replication Stress Response

Nuclear actin polymerization is essential for efficient DNA double-strand break (DSB) repair and replication stress response. Upon DNA damage, actin filaments facilitate the directed movement of DSBs and stalled replication forks toward the nuclear periphery, promoting repair through homology-directed mechanisms [\[1\]](#) [\[33\]](#). Key ABPs such as WASp, N-WASp, DIAPH1, and ARP2/3

are recruited to replication forks upon replicative stress, and their depletion significantly reduces fork velocity and impairs fork restart [1].

Genotoxic stress-induced slowing and reversal of fork growth depend on ARP2/3-mediated nuclear actin branching [1]. Inhibiting actin polymerization under stress conditions induces unrestrained fork progression and discontinuous DNA synthesis by PrimPol primase-polymerase, suggesting that nuclear actin filaments limit access of potentially dangerous replication factors to facilitate proper repair [1] [33].

Table 2: Nuclear Actin in Genome Maintenance Mechanisms

Nuclear Process	Actin Form	Key Regulatory Proteins	Functional Outcome
Chromatin Remodeling	Monomeric (within complexes)	SWI/SNF, INO80, NuA4 components	Nucleosome positioning, chromatin accessibility
3D Genome Organization	Not fully characterized	β -actin, PRC2, SWI/SNF	Chromatin compartmentalization, promoter-enhancer interactions
DNA Repair	Polymeric	ARP2/3, Formins, Myosins	Directed movement of DNA breaks, repair focus formation
Replication Stress Response	Polymeric	WASp, N-WASp, DIAPH1, ARP2/3	Fork reversal, restart, PrimPol regulation

Methodological Approaches for Studying Nuclear Actin in Disease

Experimental Models and Genetic Manipulations

The study of nuclear actin in disease contexts employs diverse experimental models, each offering distinct advantages. Mouse embryonic fibroblasts (MEFs) derived from β -actin knockout embryos have been instrumental in elucidating the role of nuclear actin in chromatin organization and transcriptional regulation [8]. These cells exhibit an intact cytoplasmic cytoskeleton due to compensatory increases in α -smooth muscle actin, allowing specific investigation of nuclear actin functions [8].

Genetic manipulation techniques include siRNA- and shRNA-mediated knockdown, CRISPR/Cas9-mediated gene editing, and overexpression of wild-type or mutant actin variants. The G13R polymerization-defective actin mutant has been particularly useful for studying actin's

polymerization-dependent functions, as it exhibits higher affinity for replication protein A (RPA) than wild-type or hyperpolymerizing S14C mutant actin [1].

Imaging and Molecular Analysis Techniques

Advanced imaging approaches have been crucial for visualizing nuclear actin dynamics. Super-resolution microscopy techniques, particularly structured illumination microscopy (SIM) with high inclined and laminated optical (HILO) illumination, have enabled visualization of nuclear actin filaments and their association with transcription machinery [34]. These methods can resolve actin-containing structures at approximately 60 nm spatial resolution, allowing identification of distinct Pol II cluster types and their association with nuclear actin [34].

Biochemical and genomic methods for studying nuclear actin include:

- **CUT&Tag assays** for mapping protein-genome interactions [34]
- **Chromatin Immunoprecipitation (ChIP)** for analyzing histone modifications like H3K27ac [8]
- **Formaldehyde crosslinking-RNA immunoprecipitation (fRIP)** for studying RNA-protein interactions [8]
- **Hi-C and related techniques** for analyzing 3D genome organization [8]
- **Activity by contact (ABC) analysis** for integrating multi-omics data to identify functional promoter-enhancer interactions [8]

Nuclear Actin in Cancer Pathogenesis

ACTN1 in Tumor Progression and Therapy Resistance

Alpha-actinin-1 (ACTN1), an actin cross-linking protein traditionally associated with cytoplasmic functions, has emerged as a significant contributor to cancer progression and therapy resistance. In head and neck squamous cell carcinoma (HNSCC), ACTN1 overexpression correlates with suboptimal response to neoadjuvant chemotherapy and reduced overall survival [35].

Mechanistically, ACTN1 activates β -catenin-mediated signaling by promoting the interaction between myosin heavy chain 9 (MYH9) and GSK-3 β , leading to ubiquitin-dependent degradation of GSK-3 β [35]. Additionally, ACTN1 interacts with integrin β 1, subsequently activating the FAK/PI3K/AKT pathway, providing an alternative mechanism for β -catenin activation [35].

Similar ACTN1-driven mechanisms operate in other malignancies. In thyroid cancer (THCA), ACTN1 is significantly upregulated and associated with tumor size, extraglandular invasion, lymph node and distant metastasis, and poor prognosis [36]. ACTN1 promotes THCA cell proliferation, cell cycle progression, migration, invasion, and epithelial-mesenchymal transition (EMT) through

activation of the PI3K/AKT/mTOR pathway [36]. In gastric cancer, ACTN1 regulates EMT and tumorigenesis via the AKT/GSK3β/β-catenin pathway [37].

Anillin (ANLN) in Transcriptional Regulation

Anillin (ANLN), a scaffold protein best known for its role in cytokinesis, also functions in the nucleus during interphase, where it regulates transcription [34]. Nuclear ANLN directly interacts with the RNA polymerase II large subunit to form transcriptional condensates, enhances initiated Pol II clustering, and promotes Pol II C-terminal domain (CTD) phase separation [34].

Short-term depletion of ANLN alters chromatin binding and enhancer-mediated transcriptional activity of Pol II. The target genes regulated by the ANLN-Pol II axis are involved in oxidoreductase activity, Wnt signaling, and cell differentiation [34]. THZ1, a super-enhancer inhibitor, specifically inhibits ANLN-Pol II clustering, target gene expression, and esophageal squamous cell carcinoma (ESCC) cell proliferation, suggesting therapeutic potential for targeting ANLN-Pol II interactions in cancer [34].

Nuclear Actin Dynamics in Cancer Cell Adaptation

Cancer cells exploit nuclear actin dynamics to adapt to therapeutic challenges. For instance, the small GTPase RhoJ is specifically upregulated in cancer cells that have undergone epithelial-mesenchymal transition (EMT), enabling them to activate DNA damage response mechanisms in response to chemotherapy [33]. RhoJ regulates nuclear actin by interacting with and modulating actin-interacting proteins, leading to enhanced DNA damage repair that can be reversed with F-actin inhibitors like latrunculin B [33].

Similarly, replication stress-induced nuclear actin polymerization facilitates the mobilization of stressed replication foci to the nuclear periphery and directs their movement along nuclear filaments with the assistance of myosin II [33]. These mechanisms promote replication stress repair and are active in xenograft tumor models following treatment with replication stress-inducing chemotherapeutics [33].

The Scientist's Toolkit: Essential Research Reagents and Methodologies

Table 3: Key Research Reagents for Nuclear Actin Studies

Reagent/Category	Specific Examples	Function/Application	Reference
Cell Models	β-actin KO MEFs, Cancer cell lines (HNSCC, THCA, ESCC)	Disease modeling, functional studies	[8] [35] [34]

Reagent/Category	Specific Examples	Function/Application	Reference
Genetic Tools	siRNA/shRNA against ACTN1, ANLN, β -actin; Mutant actin constructs (G13R, S14C)	Targeted gene manipulation, structure-function studies	[1] [35] [34]
Chemical Inhibitors	Latrunculin B (F-actin), CK-666 (ARP2/3), THZ1 (CDK7/Pol II), 740Y-P (PI3K activator)	Pathway modulation, mechanistic studies	[36] [33] [34]
Antibodies	Anti-actin, Anti-Pol II pS5, Anti-H3K27ac, Anti-ACTN1	Detection, localization, enrichment studies	[8] [34]
Imaging Tools	SIM/HILO microscopy, PLA reagents, Live-actin probes (e.g., F-tractin)	Visualization of nuclear structures and dynamics	[34]
Omics Methods	CUT&Tag, ChIP-seq, Hi-C, RNA-seq, ATAC-seq	Genome-wide mapping, chromatin analysis	[8] [34]

Concluding Perspectives and Therapeutic Implications

The emerging understanding of nuclear actin's roles in gene regulation has profound implications for understanding disease pathogenesis and developing novel therapeutic strategies. The connections between nuclear actin dysregulation and cancer progression highlight potential intervention points, particularly for treatment-resistant malignancies.

Several approaches show promise for therapeutic targeting of nuclear actin-dependent processes:

- **Small molecule inhibitors** of nuclear actin polymerization or associated regulatory proteins
- **Targeted disruption** of specific protein-protein interactions, such as those between ANLN and Pol II
- **Modulation of phase separation** processes that depend on nuclear actin
- **Exploitation of synthetic lethal relationships** that emerge in cells with nuclear actin perturbations

Future research directions should focus on elucidating the specific structural features of nuclear actin that distinguish it from cytoplasmic pools, developing more precise tools for manipulating nuclear actin without disrupting essential cytoplasmic functions, and identifying biomarkers that

predict responsiveness to therapies targeting nuclear actin networks. As our understanding of nuclear actin biology continues to mature, so too will opportunities for translating these insights into improved human health outcomes.

Conclusion

Nuclear actin emerges as a central coordinator of transcriptional regulation through multiple interconnected mechanisms: direct participation in RNA polymerase complexes, chromatin remodeling activity, regulation of transcription factor activity, and facilitation of RNA polymerase II clustering through phase separation. The development of sophisticated imaging and genetic tools has revealed nuclear actin's essential role in rapid transcriptional responses to stimuli and its importance in developmental and cellular reprogramming contexts. Validation in disease models, particularly cancer, highlights the therapeutic potential of targeting nuclear actin-mediated transcriptional regulation, with THZ1 representing a promising lead compound. Future research should focus on developing more specific nuclear actin probes, elucidating the structural basis of actin-polymerase interactions, and exploring nuclear actin's roles in additional disease contexts beyond cancer. The integration of nuclear actin biology into drug discovery pipelines represents an exciting frontier for developing novel therapeutics that modulate gene expression at its most fundamental level.

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